Generative AI Declaration

Tool: Claude (Anthropic), used through Claude Code.
Purpose: Double checking and debugging of code to ensure that experiments are runnable
Verification:

- Syntax & Import Validation: Performed a comprehensive codebase scan using Claude Code to verify that all dependencies are accounted for, relative file paths are correct, and no unresolved imports or syntax errors exist

Question A1

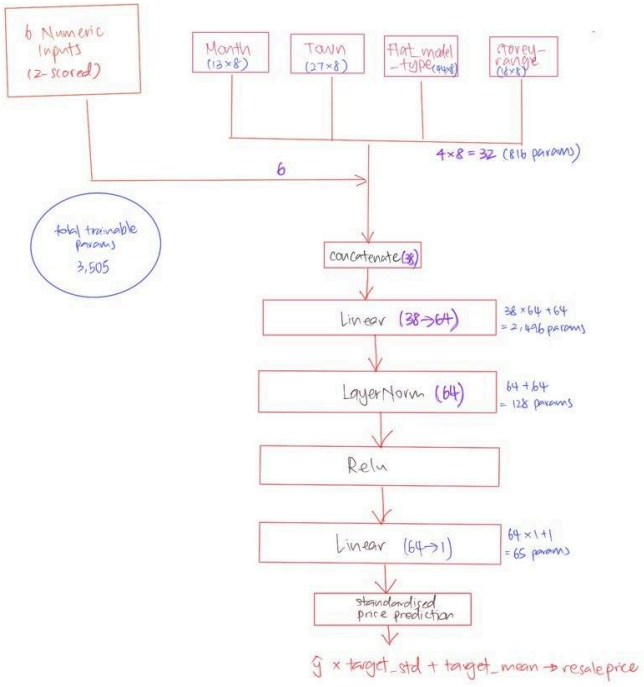


Figure 1(a) Block Diagram of baseline with no. of trainable params

Trainable parameters: The baseline model has a total of 3,505 trainable parameters - 816 in the four embedding tables ((13 + 27 + 44 + 18) × 8), 2,496 in the first linear layer (38 × 64 + 64), 128 in the LayerNorm and 65 in the output layer.

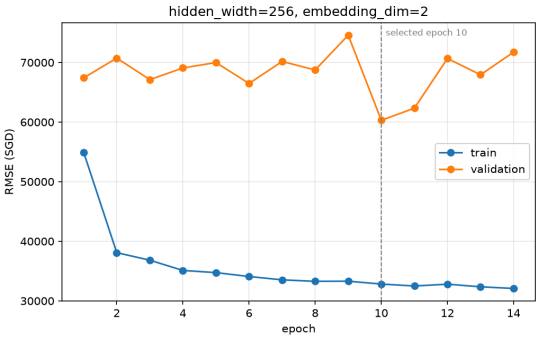


Figure 1(b) training and validation RMSE curves

h_width / e_dim	2	4	8	12	16
16	66,426	68,880	67,520	62,655	67,165
32	69,204	62,886	66,200	62,480	64,593
64	63,492	64,634	63,422	64,105	63,033
128	66,329	61,741	66,832	60,332	65,600
256	60,306	60,578	63,324	65,746	62,915

Figure 1(c) Grid Search using RMSE on all 25 trials

Wider hidden layers show lower validation RMSE, not monotonically. Embedding dimension shows no clear trend. Across 25 trials, validation RMSE ranges from 60,306 to 69,204 (sd 2,516). The top configurations, 256/2 (60,306) and 128/12 (60,332), differ by just 26 SGD and are effectively indistinguishable given seed noise. Nevertheless, 256/2 was selected for its marginally lower RMSE. Since the winner in my result sits on the edge of the searched range, the true optimum may lie outside of the grid.

Therefore, **Best validation RMSE = 60,306, Selected Config is, hidden_width = 256, embedding_dim = 2, best epoch = 10**, trainable parameters = 4,813

Retraining and Final RMSE

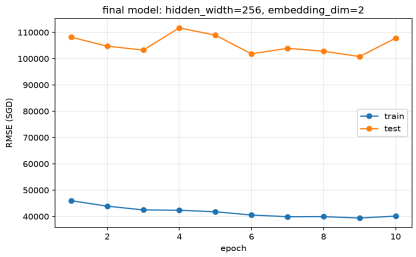


Figure 1(d) training and testing RMSE curves

Metric	Value
Train RMSE (2017-2021)	40,128
Test RMSE (2022)	**107,819**
Test R²	**0.5990**

Figure 1(e) final train and test RMSE, R²

The learning curves show no clear signs of underfitting or overfitting. From 1(d), training RMSE decreases from approximately 46k to 40k, while test RMSE remains stable at 101k–112k. From 1(b), validation RMSE similarly fluctuates between 60k and 75k. Grid search favours wider models (128–256 hidden units), suggesting additional capacity improves validation performance. With only 4,813 parameters for 87,370 training rows, severe overfitting is less likely, though it is possible. The training RMSE plateau at approximately 9% of mean training price indicates meaningful pattern learning. **Temporal distribution shift** is the likelier cause of the generalisation gap with 2022 prices average around 90k above the training-period mean, while test predictions underpredict by approximately 98k. Excluding year from the inputs prevents the model from directly accounting for rising price levels, potentially causing systematic underprediction despite capturing relative property-value differences.

Question A2

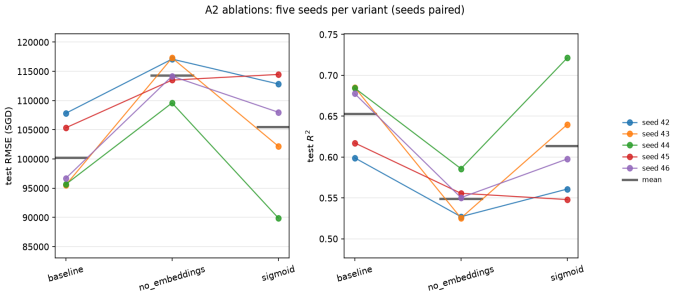


Figure 2(a) training and testing RMSE curves for baseline, no_embeddings and sigmoid

Variant	Params	Test RMSE (mean ± sd)	Test R^2 (mean ± sd)	RMSE range
Baseline (A1)	4,813	100,206 ± 5,899	0.6527 ± 0.0413	95,534 - 107,819
(a)No embeddings	2,561	114,333 ± 3,149	0.5488 ± 0.0247	109,58-117,299
(b)sigmoid	4,813	105,480 ± 9,939	0.6135 ± 0.0700	89,895 - 114,474

Figure 2(b) Average RMSE and R^2 across 5 seeds

1. Ablation (a): Removing Categorical Embeddings: Increases the mean test RMSE (+14.1%) and reduces the mean test R^2 (−0.1039). From 2(a), this degradation is consistent across all five seeds, with RMSE increasing and R^2 decreasing. Notably, The performance drop is 2.4 times the baseline's seed-to-seed standard deviation (5,899), with no distribution overlap. Even the worst baseline RMSE (107,819) outperforms the best no-embeddings RMSE (109,589), so the effect cannot be explained by initialisation noise. These results show that the categorical features carry substantial predictive information that the continuous features do not (storey_range, flat_model_type and month have no direct continuous equivalent, and the distance features only partly capture town).

2. Ablation (b): Replacing ReLU with Sigmoid: Increases the mean test RMSE (+5.26%) and mean test R^2 falls (−0.039). However, this mean penalty is smaller than the variant's own standard deviation (9,939) and its range fully spans the baseline's. In Figure 2(a), Sigmoid is worse for four of five seeds, but seed 44 improves to 89,895, the best run of any variant. On five seeds, the difference cannot be separated from initialisation noise. The larger standard deviations (RMSE: 5,899 → 9,939; R^2: 0.041 → 0.070) also indicate less stable performance. With identical parameter counts, capacity is not the cause. Instead, LayerNorm may keep pre-activations near zero, where Sigmoid behaves almost linearly and provides less nonlinearity than ReLU.

Overall, categorical embeddings have a substantial, statistically significant impact on performance, whereas Sigmoid activation causes only a small, inconsistent penalty within seed noise.

Question A3

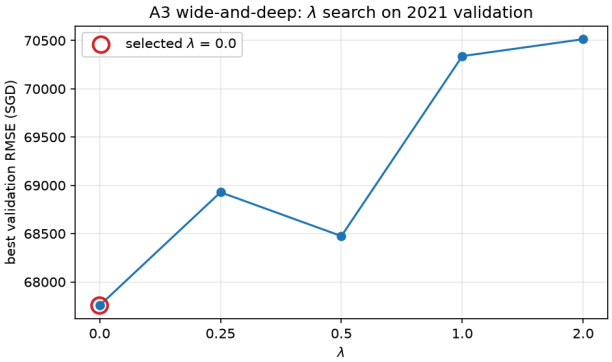


Figure 3(a) RMSE on λ in {0, 0.25, 0.5,1,2}

From Figure 3(a), **Selected λ = 0.**

RMSE range (67,756 - 70,514) = 2759

Since λ=0 renders the wide branch inert, it will contribute nothing to the output and receives zero gradient As such, the best non-zero value (λ=0.5) was additionally evaluated in Figure 3(b) so that a genuine wide-and-deep model is measured. [This is reported as a supplementary diagnostic, not a re-selection]

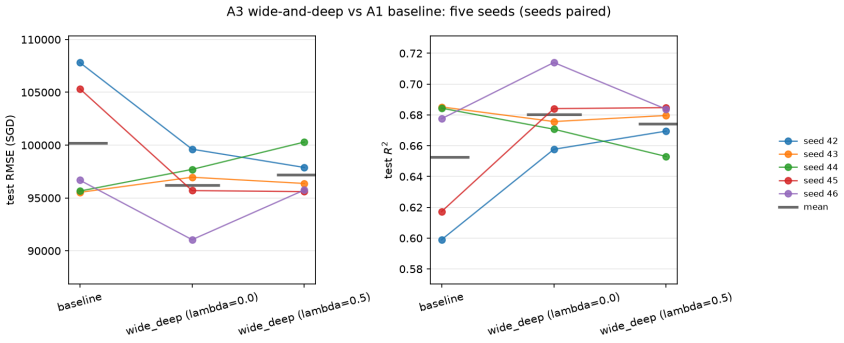


Figure 3(b) RMSE and R^2 for baseline, λ=0.0, λ=0.5

Variant	Params	Test RMSE (mean ± sd)	Test R^2 (mean ± sd)
Baseline (A1)	4,813	100,206 ± 5,899	0.6527 ± 0.0413
Wide-and-Deep, λ = 0.0	4,828	96,207 ± 3,212	0.6804 ± 0.0211
Wide-and-Deep, λ = 0.5	4,828	97,182 ± 1,960	0.6741 ± 0.0132

Figure 3(c) Average RMSE and R^2 for 5 seeds

Both wide-and-deep variants appear slightly better than the baseline from Figure 3(b) and 3(c), but this does not show that the architecture helps. In the original wide-and-deep formulation, the wide branch receives sparse cross-product features to memorise specific feature combinations, while the deep branch generalises from broader patterns. Here, both branches receive the same 14 dense features, and the MLP already begins with Linear(14, 256). The wide branch therefore adds only 15 parameters and a direct linear path that the MLP can already represent, so little effect on generalisation is expected. Its possible effect is on optimisation by offering an alternative gradient path, although a large λ may amplify an unhelpful linear signal early in training.

The evidence is consistent with this mechanism. The validation RMSE across λ values spans only 2,759 (67,756 at λ = 0 to 70,514 at λ = 2), smaller than the baseline's seed-to-seed standard deviation of 5,899 in A2. The search selected λ = 0, and the two largest values (λ = 1, 2) produced the worst validation scores, so the wide branch offers no consistent benefit.

A key confound is batch ordering. Although the λ=0 configuration is mathematically equivalent to the baseline, the extra parameters alter the random-number sequence and hence batch order. The resulting differences in mean test RMSE (3,999) and R^2 (0.028) exceed the λ=0.5 variant's apparent RMSE improvement of 3,024, making this improvement unreliable evidence of architectural benefit. The λ search also used a single seed (42) on the validation year, so the ranking of λ values is itself subject to seed noise.

Question A4

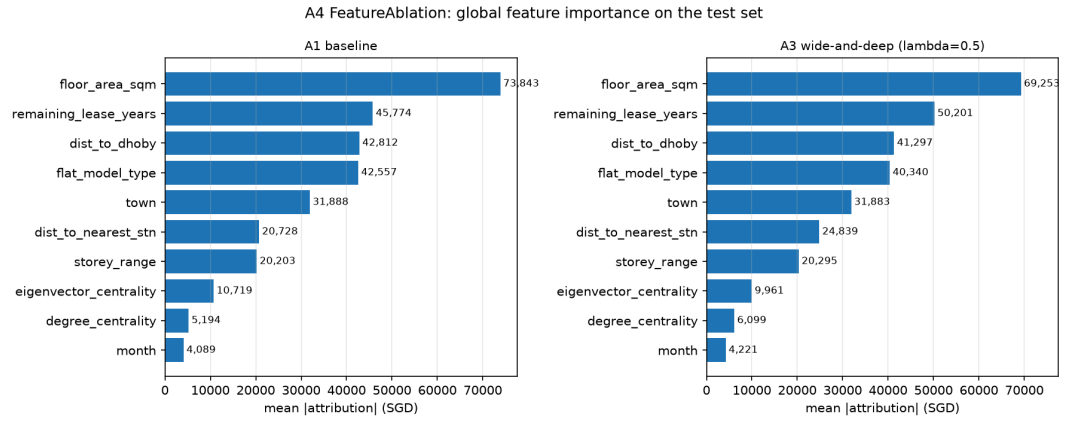


Figure 4 Ranking of features for baseline and λ=0.5 as a supplementary diagnostic

Top three: *floor_area_sqm, remaining_lease_years, dist_to_dhoby.*

Both models rank all ten features identically but differ in predicting the importance of each feature, with the largest relative gap to be “dist_to_nearest_stn” (+20%). This is consistent with A3 whereby the wide branch receives the same dense input the MLP already has, adds no new capability, and so does not change what the model relies on. There is also no preference for categorical or continuous feature type.

The ranking is economically sensible with flat size dominating, then remaining lease (the 99-year decay), then distance to the city centre. The two graph-centrality measures rank lowest of the continuous features, and month is last, which makes sense since it encodes only month-of-year with no time trend.

Limitation: FeatureAblation’s inability to capture feature interactions. It perturbs one feature at a time, so its scores are decomposable contributions only if the model is additive in those features, which it is not. If it were, the joint ablation of dist_to_dhoby and eigenvector_centrality would equal the sum of their individual attributions for every flat, but instead the two differ by 12,123 SGD per flat on average (28.7% of the joint attribution magnitude of 42,303 SGD, with a maximum discrepancy of 222,656 SGD). That average discrepancy exceeds eigenvector_centrality's entire individual score of 10,719 SGD. The two features are negatively correlated ($r = -0.57$), which is consistent with their contributions partly offsetting. Individual scores should therefore be read as ordinal indicators rather than exact additive contributions to the predicted price.